

Linear algebra is a study of algebraic mathematical structures called ‘vector spaces’ and their structure preserving maps ‘linear transformations’, we shall define these terms precisely later. Linear algebra is also applicable in to wide range of fields: physics, economics, computer science, chemistry etc.

In MAM2000W, we hope this module will help in developing your thinking about mathematics more ‘abstractly’ (in preparation for 2IA) and developing ideas useful in solving some (systems of) differential equations (for 2DE). Now, we begin by recalling your first year algebra of matrices and systems of linear equations.

The algebra of matrices

In your first year mathematics, you may have defined *matrices* (over real or complex numbers) as follows:

Definition 1. Let m and n be positive integers. An $m \times n$ matrix A is a rectangular array of numbers in m rows and n columns

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

here each entry a_{ij} denotes a number in the i -th row and j -th column of matrix A and $m \times n$ is called the size of matrix A .

Convention 1. We shall denote matrices by uppercase letters and their entries by lowercase letters. We shall also simplify our notation and denote the above matrix by $A = (a_{ij})$.

We call $1 \times n$ matrices row vectors and $m \times 1$ matrices column vectors.

Examples Below are examples of matrices of sizes 2×3 , 2×3 , 1×4 and 1×1 , respectively.

$$\begin{pmatrix} 0 & 0 & e \\ 2 & 7 & 6 \end{pmatrix}, \begin{pmatrix} 0 & 8 & 1 \\ 2 & 7 & 6 \end{pmatrix}, (7 \quad 2+i \quad 3), 7$$

Definition 2. Let $A = (a_{ij})$ and $B = (b_{ij})$ be $m \times n$ and $p \times q$ matrices, respectively. Then A and B are equal precisely when they have the same size and their corresponding entries equal, that is, $A = B$ if and only if $m = p, n = q$ and entries $a_{ij} = b_{ij}$ for each $1 \leq i \leq m$ and $1 \leq j \leq n$.

All the 4 matrices given above are not equal, the first two have the same size but their corresponding entries in the first row and second column are not equal.

Definition 3. Let $A = (a_{ij})$ and $B = (b_{ij})$ be $m \times n$ matrices. We define their addition $A + B$ componentwise, that is, by adding corresponding entries. Using our notation $A + B = (a_{ij} + b_{ij})$.

Example

$$\begin{pmatrix} 0 & 0 & e \\ 2 & 7 & 6 \end{pmatrix} + \begin{pmatrix} 0 & 10 & \pi \\ 3 & 0 & 12 \end{pmatrix} = \begin{pmatrix} 0 & 10 & e + \pi \\ 5 & 7 & 18 \end{pmatrix}$$

Let $\mathbf{0}$ denote the $m \times n$ matrix with all its entries 0 and given an $m \times n$ matrix $A = (a_{ij})$, $-A$ the $m \times n$ matrix with its corresponding entries additive inverses of entries of A , that is, $-A = (-a_{ij})$, then this componentwise addition will immediately give the following properties.

Proposition 1. *Let A, B, C be $n \times m$ matrices, then we have*

- $A + (B + C) = (A + B) + C$; (*associativity*)
- $A + \mathbf{0} = A = \mathbf{0} + A$; (*identity laws*)
- $A + (-A) = \mathbf{0} = -A + A$; (*inverse laws*)
- $A + B = B + A$. (*commutativity*)

Given that 4 holds, that is commutativity, in the previous proposition, we may omit the second equalities of the identity and inverse laws, but we chose to include them.

We have more than just addition. Let us begin by multiplication by *scalars*

Definition 4. *Let $A = (a_{ij})$ be an $m \times n$ matrix and λ a number. We define scalar multiplication λA by (left) multiplying each entry of matrix A by λ , that is, $\lambda A = (\lambda a_{ij})$*

Example

$$5 \begin{pmatrix} 0 & 10 & 7 \\ 5 & 7 & 18 \end{pmatrix} = \begin{pmatrix} 0 & 50 & 35 \\ 25 & 35 & 90 \end{pmatrix}$$

Again, usual properties of addition and multiplication of numbers immediately gives us the following properties.

Proposition 2. *Let A and B be $m \times n$ matrices, λ and μ numbers, then we have*

- $1A = A$;
- $\lambda(\mu A) = (\lambda\mu)A$;
- $(\lambda + \mu)A = \lambda A + \mu A$;
- $\lambda(A + B) = \lambda A + \lambda B$.

Other properties like $0A = \mathbf{0}$ and $(-1)A = -A$, though also follow immediately from properties of numbers and the scalar multiplication definition, they can also be deduced from the above properties.

In the next lecture we shall begin by revisiting the matrix multiplication and its properties...